

RESEARCH PRESENTATION

KnowSelf

Agentic Knowledgeable Self-Awareness for LLM-Based Agents

Teaching LLM Agents When to Think, Reflect, or Seek Knowledge

Enabling selective, situational knowledge utilization through metacognitive self-awareness

 **Fast Thinking**

Direct action generation

 **Slow Thinking**

Self-correction via reflection

 **Knowledgeable Thinking**

External knowledge retrieval

ML Researchers & Graduate Students | April 2026

Source Paper

arxiv.org/abs/2504.03553

Llama-8B backbone on ALFWorld

4.33% accuracy · only 15% knowledge

Gemma-2B on ALFWorld

9.85% accuracy · only 26% knowledge

Current Agent Training Uses "Flood Irrigation" — Indiscriminate Knowledge Injection

Current agents blindly inject knowledge at every step — leading to fragile planning and unnecessarily high inference cost

TRAJECTORY IMITATION

e.g., ReAct

Copies gold actions step-by-step regardless of difficulty

→ No situational judgment; same treatment for easy & hard steps

TRIAL-AND-ERROR REFINEMENT

e.g., Reflexion

Applies external feedback uniformly at every planning step

→ Unnecessary reflection even on trivially correct steps

KNOWLEDGE-AUGMENTED PLANNING

e.g., KnowAgent, WKM

Injects 100% domain knowledge regardless of actual need

→ High inference cost; indiscriminate injection can hurt performance

⚠ All three use a "flood irrigation" approach — treat every step identically

Shared Limitations of All Three Paradigms

- Fragile planning — collapses on unexpected inputs
- High inference cost from 100% knowledge queries
- No situational judgment at each decision step
- Pattern-fitting, not principled decision-making
- Unconscious imitation — not self-aware reasoning



Humans Possess Situational Self-Awareness

Humans assess each situation before deciding how to proceed:

✓ "I know this — act directly" → Fast

✓ "I made an error — let me reflect" → Slow

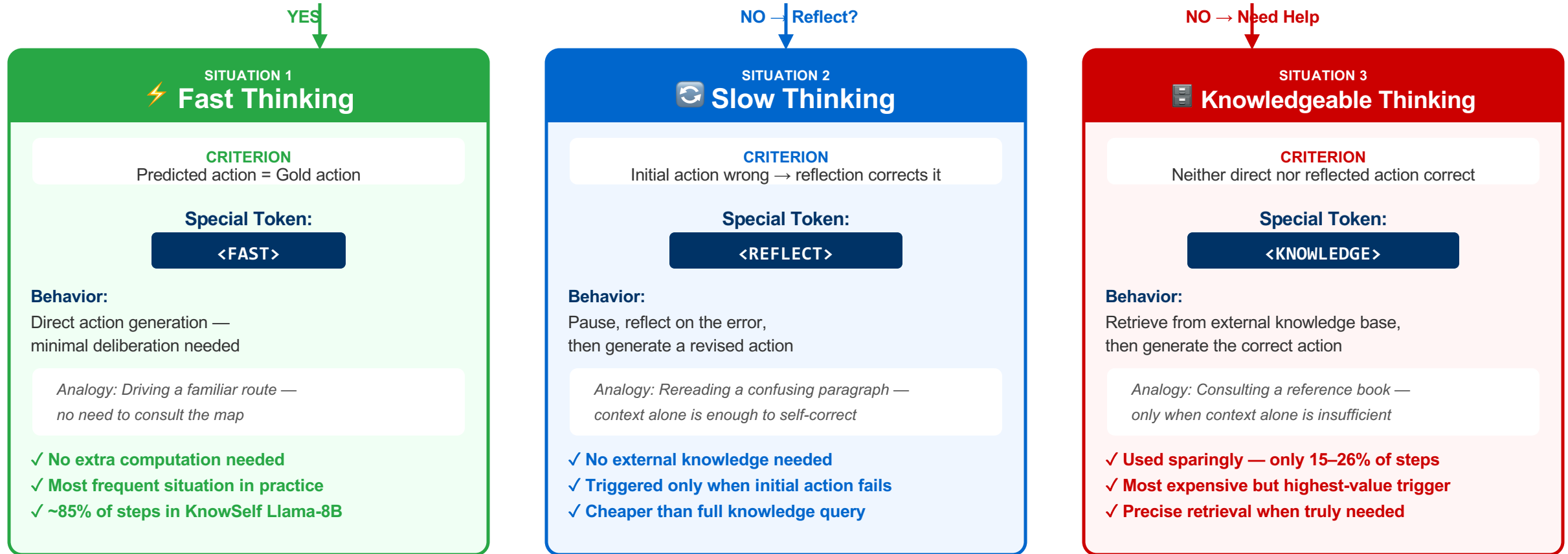
✓ "I need more information — seek help" → Knowledgeable

KnowSelf brings this metacognitive ability to LLM agents

Three Situational Modes: Fast Thinking, Slow Thinking, and Knowledgeable Thinking

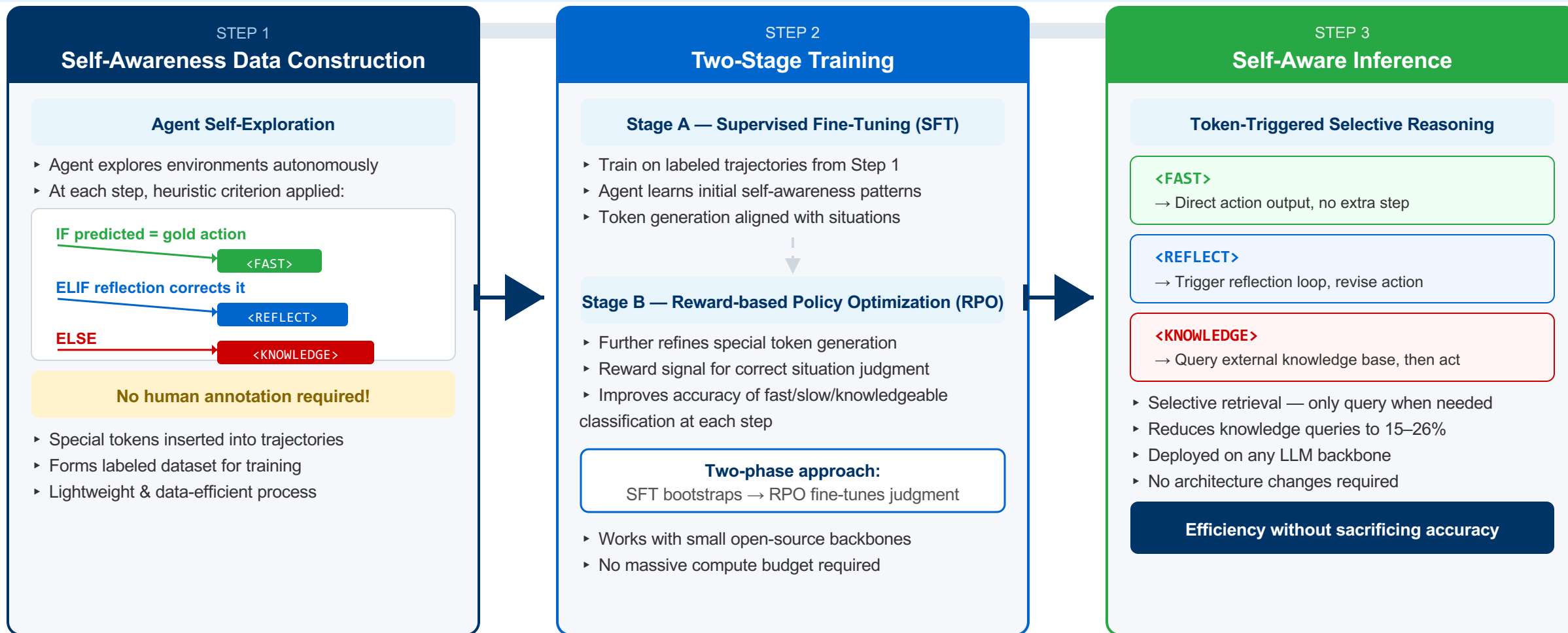
KnowSelf classifies every decision step into one of three cognitive situations — enabling selective resource allocation

At each planning step, the agent asks: "Can I produce the correct action?"



KnowSelf Framework: Data Construction → Two-Stage Training → Self-Aware Inference

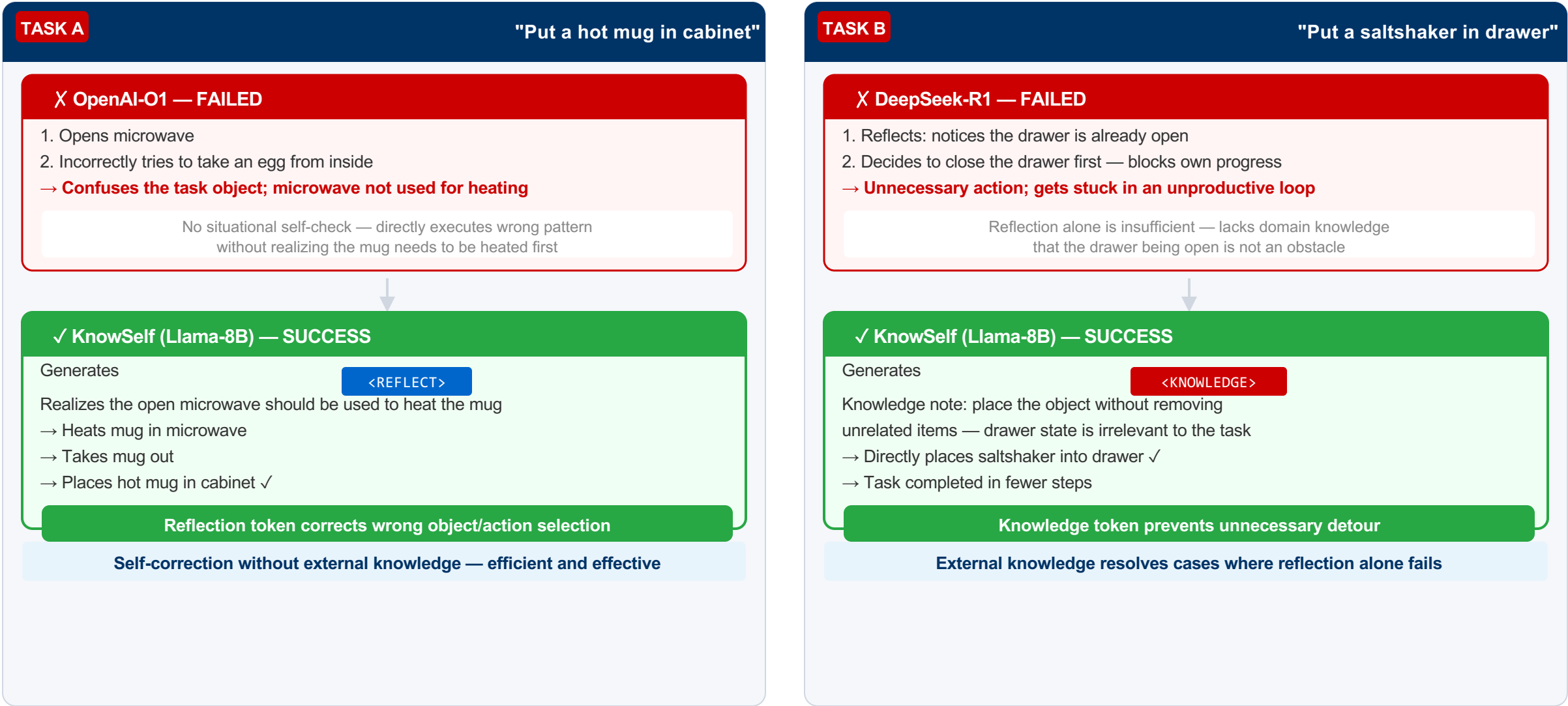
A three-stage pipeline: self-labeled exploration data → two-stage training → special token-driven selective inference



Source: arxiv.org/abs/2504.03553, Figure 3 — KnowSelf framework overview

KnowSelf Correctly Heats a Mug and Places a Saltshaker Where Baselines Fail

KnowSelf uses the right cognitive mode — reflection or knowledge — where frontier models apply the wrong strategy



Source: arxiv.org/abs/2504.03553, Figure 1 — Qualitative comparison: KnowSelf vs. OpenAI-O1 and DeepSeek-R1

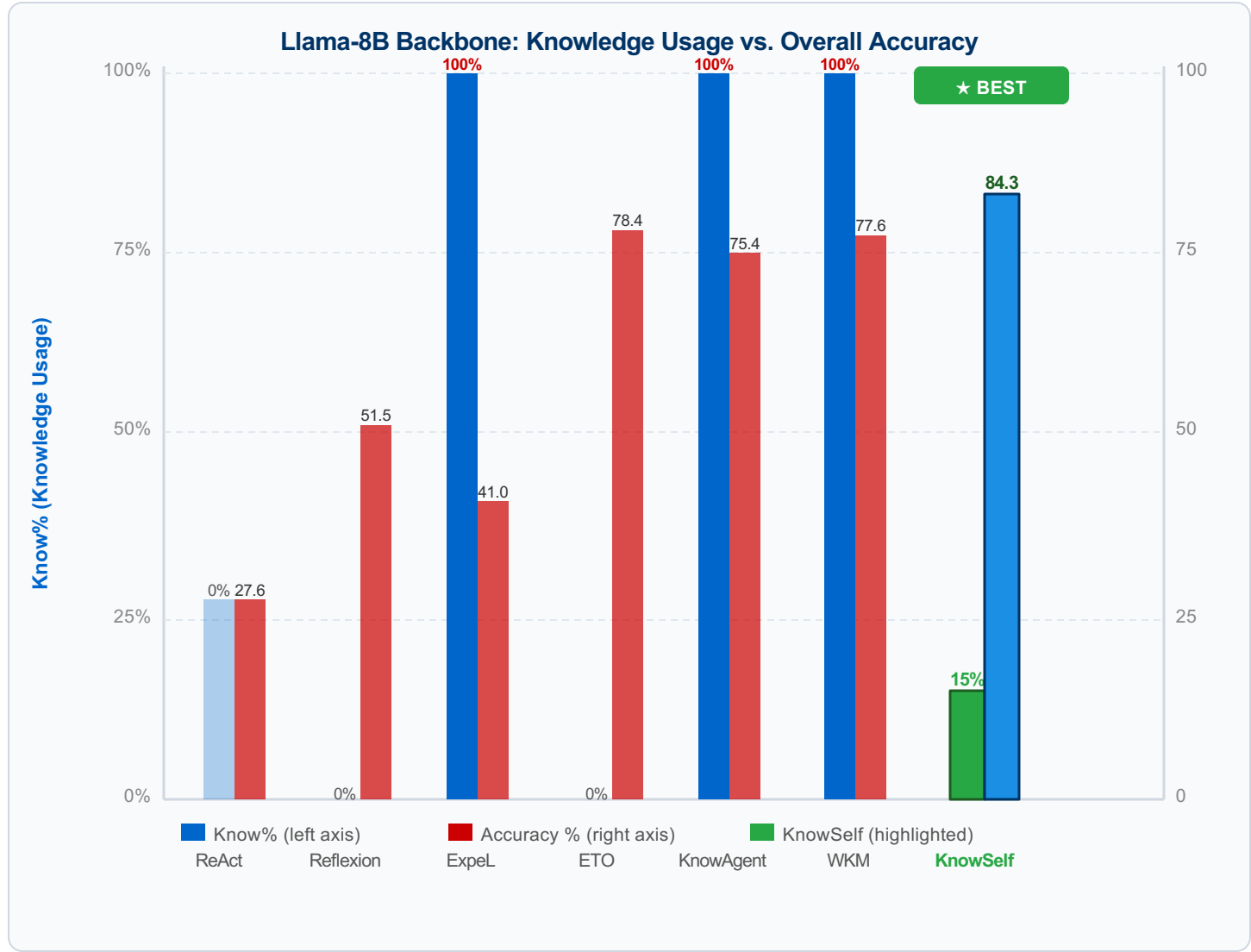
KnowSelf Achieves 84.33% on ALFWorld with Llama-8B Using Only 15% Knowledge

KnowSelf (green rows) outperforms all same-backbone baselines — including 100%-knowledge methods — using only 15–26% knowledge

Backbone	Method	Know%	Put	Clean	Heat	Cool	Examine	Put Two	All (%)
GPT-4o	ReAct	0%	83.33	74.19	69.57	85.71	77.78	64.71	76.12
GPT-4o	Reflexion	0%	100.00	87.10	73.91	90.48	83.33	70.59	85.07
GPT-4o	ExpeL	100%	95.83	83.87	69.57	80.95	88.89	52.94	79.85
Llama-8B	ReAct	0%	33.33	3.23	0.00	57.14	66.67	23.53	27.61
Llama-8B	Reflexion	0%	62.96	22.73	5.88	64.29	86.36	50.00	51.49
Llama-8B	ExpeL	100%	83.33	32.26	30.43	23.81	55.56	17.65	41.04
Llama-8B	ETO	0%	91.67	70.59	82.61	61.90	88.89	64.71	78.36
Llama-8B	KnowAgent	100%	87.50	93.55	65.22	66.67	61.11	64.71	75.37
Llama-8B	WKM	100%	95.83	87.10	86.96	61.90	66.67	52.94	77.61
Llama-8B	KnowSelf ★	15.01%	91.67	87.10	91.30	85.71	77.78	64.71	84.33 ★
Gemma-2B	ReAct	0%	0.00	9.68	0.00	4.76	44.44	0.00	8.96
Gemma-2B	ETO	0%	91.67	83.87	78.26	52.38	77.78	29.41	71.64
Gemma-2B	KnowAgent	100%	91.67	90.32	69.57	71.43	66.67	41.18	73.88
Gemma-2B	WKM	100%	91.67	87.10	78.26	71.43	61.11	52.94	76.12
Gemma-2B	KnowSelf ★	26.41%	87.50	93.55	73.91	76.19	83.33	52.94	79.85 ★

Minimal Knowledge, Maximum Impact: KnowSelf's Efficiency Advantage

15–26% knowledge queries delivers higher accuracy than 100% injection — selective beats blanket



KnowSelf Llama-8B Overall

84.33%

★ BEST

Outperforms all Llama-8B baselines

Knowledge Queries Used

15%

vs. 100% for competitors

KnowSelf Gemma-2B

79.85%

= GPT-4o ExpeL (100% know)

Key Efficiency Insights

- ▶ **Selective beats blanket injection**
15% queries, higher accuracy than 100% methods
- ▶ **Scales down effectively to small models**
Gemma-2B matches GPT-4o ExpeL with 26% queries
- ▶ **100% knowledge can hurt performance**
ExpeL (Llama-8B, 100%): 41.04% vs 84.33% KnowSelf
- ▶ **Inference cost dramatically reduced**
6–7× fewer knowledge queries at deployment time
- ▶ **Special token overhead is negligible**
Lightweight heuristic labeling, no human annotation

Key Takeaways: Self-Awareness Is the Missing Piece in Agent Planning

KnowSelf demonstrates that agents can learn WHEN to think, reflect, or seek knowledge — selectively and efficiently

1 Flood irrigation is suboptimal

Indiscriminate knowledge injection at every step is both wasteful and harmful. Methods using 100% knowledge queries can underperform selective approaches.

ExpeL (Llama-8B, 100%): 41.04% vs KnowSelf: 84.33%

2 Lightweight self-awareness labeling

Agents learn situational self-awareness via heuristic labeling on self-explored trajectories. No expensive human annotation or massive feedback needed.

SFT + RPO on self-explored trajectories only

3 Beats strong baselines including GPT-4o

KnowSelf with Llama-8B surpasses all Llama-8B baselines and even frontier model results in this planning benchmark. Small model + good strategy > large model.

84.33% vs GPT-4o ReAct 76.12% / ExpeL 79.85%

4 Dramatically reduced inference cost

Only 15–26% knowledge queries needed vs. 100% for competing methods — a 4–7× reduction in retrieval operations without sacrificing (and often improving) accuracy.

15.01% (Llama-8B) · 26.41% (Gemma-2B)

5 Generalizes across model scales

Even Gemma-2B (smaller) reaches 79.85%, matching GPT-4o ExpeL and exceeding all other Gemma-2B baselines. The paradigm scales down effectively to small models.

Gemma-2B KnowSelf = GPT-4o ExpeL (100% know)

Future Directions ▶ Extend to multi-modal agents | Broader planning benchmarks beyond ALFWorld | Dynamic knowledge base construction | Richer reward signals for RPO training

Source: arxiv.org/abs/2504.03553 — KnowSelf: Agentic Knowledgeable Self-Awareness for LLM-Based Agents

Thank you — Questions?

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